# Dominic Bair

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github.com/DothWalrus

## Research Interests

My research lies at the intersection of geometric group theory and probability, with emphasis on boundary measures, regular languages, random metrics, and relatively hyperbolic groups. I study how finite measures and random processes encode the asymptotic geometry of groups, including through Patterson–Sullivan theory and first-passage percolation.

## Education

2022 – Present **Ph.D. candidate, Mathematics,** University of Tennessee, Knoxville.
Advisors: Yulan Qing and Jan Rosiński.
Expected completion: May 2027.

2020 – 2022 **M.S. Mathematics,** Montana State University.
Thesis: *DVM: A deep learning algorithm for minimizing functionals*.

2016 – 2020 **B.S. Mathematics; Minor in Physics,** Montana State University.

## Publications and Preprints

[1] **D. Bair**, *Ordered frequency in relatively hyperbolic groups*, Preprint, Oct. 19, 2026. arXiv: 2610.27268 [math.GR].

[2] **D. Bair**, S. Jana, and Y. Qing, *First-passage percolation, non-positive curvature, and radial maps*, Preprint, 2025. arXiv: 2512.05720 [math.MG].

[3] C. Potts, J. Schearer, T. A. Sebrell, **D. Bair**, B. Ayler, J. Love, J. Dankoff, P. R. Harris, D. Zosso, and D. Bimczok, "MNPmApp: An image analysis tool to quantify mononuclear phagocyte distribution in mucosal tissues," *Cytometry Part A*, vol. 101, no. 12, pp. 1012–1026, 2022. DOI: 10.1002/cyto.a.24657.

## Academic Appointments

2022 – Present **Graduate Teaching Assistant,** University of Tennessee, Knoxville.

2020 – 2022 **Graduate Teaching Assistant,** Montana State University.

2019 – 2020 **Undergraduate Research Assistant,** Department of Mathematical Sciences, Montana State University.

2018 – 2019 **Undergraduate Research Assistant,** Department of Physics, Montana State University.

## Teaching Experience

UTK Statistical Reasoning; Basic Calculus; Mathematical Reasoning; College Algebra.

MSU Calculus I; Precalculus; Survey of Calculus; College Algebra.

## Selected Talks

2026 *Ordered frequency in relatively hyperbolic groups*, AMS Eastern Sectional Meeting: Special Session on Ergodic Geometry.

2026 *First-passage percolation, non-positive curvature, and radial maps*, YGGT XIV.

2026 *Graphons and random graphs*, UTK Graduate Student Seminar.

2025 *Introduction to first-passage percolation and subadditive processes*, UTK Probability Seminar.

2025 *First-passage percolation on hyperbolic graphs: bi-infinite geodesics, radial maps, and infinitely many thick triangles*, UTK Topology Seminar.

2020 *Spatial statistics in histology images*, SIAM Northern States Section Student Chapters Conference.

2020 *Spatial statistics in histology images*, Data Science and Image Analysis Conference of the Pacific Northwest.

## Additional Projects

2026 **Quanto Options,** Erdős Institute Quantitative Finance Bootcamp.
Joint work with Kevin Dao, Chloe Ireland, Ethan Katz, and Rahul Pradeep.
Developed stochastic and numerical models for pricing and hedging quanto options, whose underlying assets and payouts are denominated in different currencies.

2022 **DVM: A deep learning algorithm for minimizing functionals,** M.S. thesis, Montana State University.
Developed a neural-network method for approximating minimizers of variational problems, with applications to partial differential equations.

## Honors and Awards

2026 **Graduate Student Academic Achievement Award,** University of Tennessee, Knoxville.

2022-2026 **Departmental Fellowship,** University of Tennessee, Knoxville.

2020 **NSF and AWM Travel Grant,** Data Science and Image Analysis Conference of the Pacific Northwest.

2019 – 2020 **INBRE Grant,** support for research in spatial statistics in histology images.

2016 – 2020 **Premier Scholarship,** Montana State University.

2019 **Superior Achievement in Mathematics,** Pi Mu Epsilon.

## Service and Leadership

2026 – Present **Senator,** Research and Assessment Committee, UTK Graduate Student Senate.

2026 – Present **Leader,** UTK Mathematics Graduate Student Seminar.

## Technical Skills

Programming Python, MATLAB, Julia, Git, LaTeX.

Methods Numerical simulation, statistical computing, scientific software development, mathematical modeling, agentic workflow.